

Mitigating Tail-Risk in Institutional Endowments: A Comparative Stochastic Analysis of Hybrid Spending Rules under Non-Gaussian Market Regimes

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Problem Statement and Objectives

The fundamental challenge regarding institutional endowment management is the “intergenerational equity” paradox where institutions both mandate a providing of significant annual distributions in current operations while also simultaneously preserving the real, inflation adjusted value of the principle for future generations. Traditional endowment models often rely on “Normal distribution,” also known as a bell curve, which statistically underestimates the frequency and severity of market “shocks” or black swan events. This research project addresses this issue by developing a novel algorithm that aims to quantify resilience of large-scale institutional endowment by simulating complex market environments characterized by alternative “Fat-tail” student-t distributions and path-dependent spending logic.

KEYWORDS

Capital Preservation, Conditional Value-at-Risk (CVaR), Endowment Management, Fat-Tail Distributions, Intergenerational Equity, Jump-Diffusion Models, Monte Carlo Simulation, Path-Dependent Spending Rules, Stochastic Differential Equations (SDE), Yale Hybrid Rule

Mathematical Framework & Stochastic Engine

The core of this research is a custom-built Monte Carlo simulation engine, which unlike standard geometric brownian motion models, utilizes:

Non-Gaussian Shock Integration

Using a Student-t Distribution with 3 degrees of freedom (df=3), the algorithm employs “Leptokurtic” behavior, simulating a market where extreme events, such as three or more shifts in standard deviation, occur more frequently than in normal distributions.

Jump Diffusion Equation

By applying shocks on a daily timestep (dt=1/252), the model captures the “volatility drag” that occurs in real-world trading, providing a more in-depth view of portfolio decay than annual models. In essence, the equation accounts for significantly more dynamic events in the stock markets, often referred to as “jumps”.

The Governing Stochastic Differential Equation (SDE)

The model solves the following stochastic differential equation for the portfolio value V :

$$dV_t = V_t ((r - f)dt + \sigma dZ_t) - S_t dt + F_t dt$$

where r is expected return, f is management fees, dZ_t is the stochastic shock, S_t is the spending draw, and F_t is the fundraising inflow.

Comparative Analysis of Spending Rules

A primary focus of this study also involves the comparison between the Fixed Percentage Rule and the Yale Hybrid (Smoothing) Rule. The Yale Hybrid Rule involves a path-dependent algorithm that calculates spending as a weighted average with 80% weight on the previous year’s inflation-adjusted budget and 20% weight on the current market value with a 4.5% target rate. The conventional spending equation is modified accordingly: $S_{t+1} = w(\sigma_t) \cdot [S_t(1 + \pi)] + (1 - w(\sigma_t)) \cdot [r \cdot V_t]$. As seen from the equation, the weight is dynamically defined as a function instead of having a fixed constant value. The research moreover hypothesizes that, for the algorithm, the Yale Rule provides lower budget volatility, which could in turn lead to a higher Principal Erosion risk during prolonged “bear” markets due to its slow adjustment to falling asset prices.

Methodology & Algorithmic Implementations

The algorithm was deployed and executed through a web-based “Risk Suite” using Streamlit Python, allowing for real-time sensitivity analyses to simultaneously occur. Ultimately, the algorithm allowed for:

Dynamic Shock Testing

Users can toggle “Fat-Tail” shocks to observe the immediate impact on Value-at-Risk (VaR).

Fundraising Mitigation

The model also integrated a novel “Fundraising as % of Draw” slider. This simulates a “Self-Correcting Loop feedback” where philanthropic or sponsored effects scale with operational needs, offering a buffer against the Probability of Erosion.

Capital Project Stress-Testing

A “Shock Withdrawal” feature was integrated to model the impact of large, one-time capital expenditures (e.g., \$50M for a new facility) on the long-term survival of the principal.

Quantitative Stress Testing & Findings

The simulation runs (5,000 iterations over a 15 year horizon) yielded several critical insights that are applicable to several institutions for risk assessment and management:

The Scale-of-Loss Paradox

As fundraising increases, the Probability of Erosion significantly reduces. However, the 95% VaR increases, demonstrating that as the endowment grows, the absolute “dollar-at-risk” scales linearly with the portfolio value, even as the safety of the institution’s “floor” improves.

Smoothing Risk

The Yale Hybrid Rule effectively “smooths” the school’s budget, but in 54% of “Tail-Risk” scenarios, it creates a “lag effect” where the endowment continues to spend at high levels while the principal is rapidly depreciating, requiring higher fundraising interventions to remain stable.

Tail-Risk Quantification

Using the Student-t distribution model, the model runs risk assessments to find the Value-at-Risk (VaR) and Conditional Value-at-Risk (CoVaR) at 95% tail risk. This permitted the model to account for market “shocks” that are 22% worse than algorithms implementing normal distribution (bell curves). Such statistics indicated that conventional risk assessing algorithms substantially underestimate the severity of worst-case risk

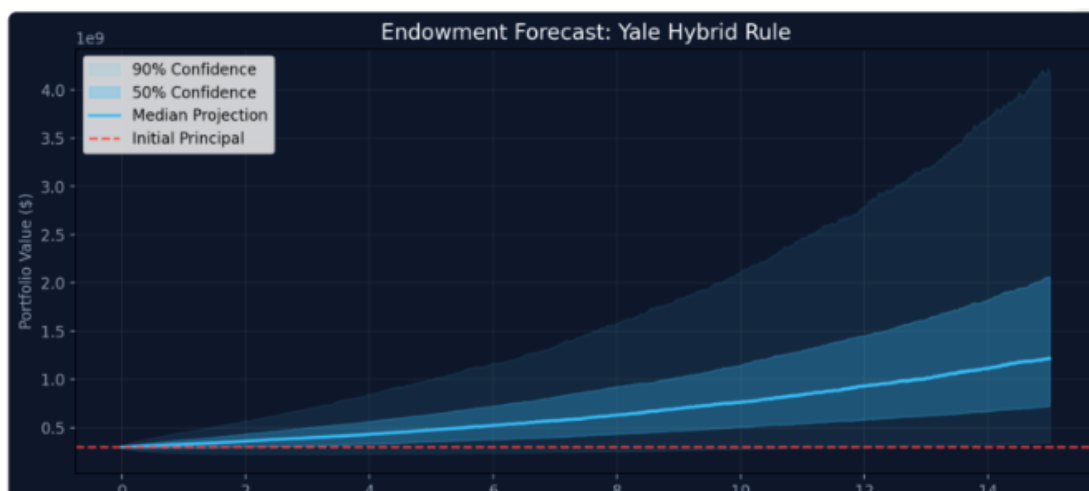


Fig 1.1

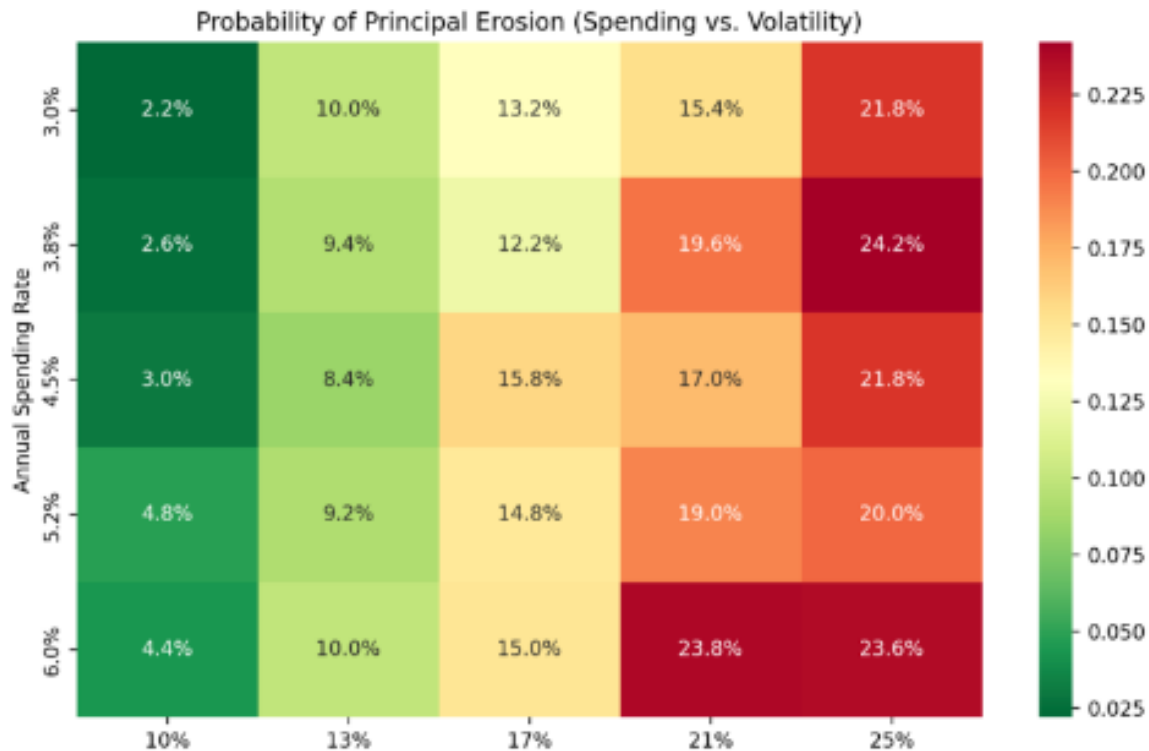


Table 1.1

Conclusion & Institutional Application

The novel algorithm developed through this research project concludes that institutional sustainability is not merely a product of market returns, but of the mathematical relationship between consumption inertia and market volatility. For an institution like Choate, the research suggests that a Hybrid spending rule is only sustainable when coupled with a robust fundraising “buffer” that scales alongside the endowment draw. This project provides a transparent, data-driven framework for fiduciaries to visualize the “trade-off” between current operational ambition and future financial security.

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